



Task 1.1

prior: $f \sim \mathcal{N}(0, k)$

2.1

Squared kernel is defined as:

$$k(x, x') = b^2 \exp\left(-\frac{\|x - x'\|_2^2}{2b^2}\right)$$

$$b_1(x, x') = 2 \exp\left(-\frac{1}{8} \|x - x'\|_2^2\right)$$

$$b = \sqrt{2}$$

$$b = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}} = \frac{1}{2 \cdot 2^{\frac{1}{2}}} = \frac{1}{2 \cdot 2^{\frac{1}{2}}}$$

2.2

$$b_2(x, x') = \exp\left(-\frac{1}{2} \|x - x'\|_2\right) + 2 \quad (8)$$

2.4 Determine analytical prior variance of $f(x) \sim GP(0, b_2(x, x')) \forall x \in \mathbb{R}$ given by kernel in eq. 8- Since we have a prior, we set $x=0 \wedge x'=0$.

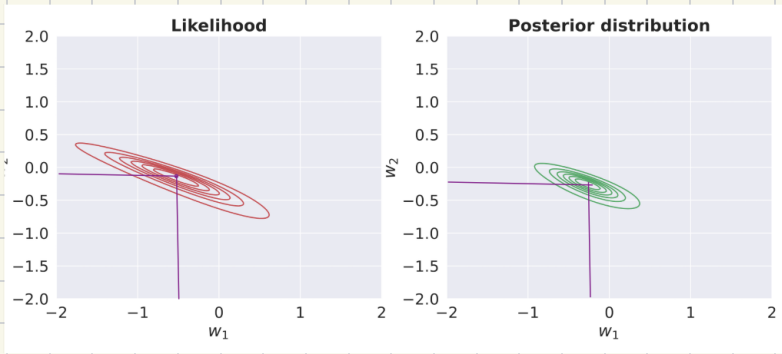
$$\text{Var}[f(x)] = b_2(0, 0) = \exp\left(-\frac{1}{2} \|0 - 0\|_2\right) + 2$$

$$= \exp\left(-\frac{1}{2} \cdot 0\right) + 2$$

$$= 1 + 2$$

$$= \underline{\underline{3}}$$

4.1



$$W_{MLE} w_1, w_2 = (-0.5, -0.2)$$

$$W_{MAP} w_1, w_2 = (-0.2, -0.2)$$

4.2 Compute the prior mean of $\mu(x^*)$ for $x^* = 0$

- the prior is $w | \alpha \sim \mathcal{N}(0, \alpha^{-1} I)$, $\alpha = 8$

$\therefore$ Prior mean of the weights: $E[w] = 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

The prior mean of the Poisson rate $\mu(x) = \exp(3 + w_1 x + w_2 x^2)$

then $a(x) = \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix}$, $\Sigma = \alpha^{-1} I$

- we use the Gaussian moment formula $E[\exp(a^T w)] = \exp(\frac{1}{2})$

5.2

$$\log p(w, y) = \log \mathcal{N}(y | e^x, 1) + \log \mathcal{N}(w_{\text{map}} | 0.1)$$

evaluating at $w_{\text{map}} = 1.293404$ and $y = 5$

$$\log p(w_{\text{map}}, 5) = \log \mathcal{N}(y | e^{w_{\text{map}}}, 1) + \log \mathcal{N}(w_{\text{map}} | 0.1) = -3.59$$